MSC MARRIAGE AND FAMILY THERAPY

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MFT 705

Research Methods in Marriage and Family Therapy

14 teaching weeks · 13 practicals · 2 figures

SEMESTER ONE

1.71 units · Prerequisite: MFT 701

> “You will become a consumer who can read a paper in ten minutes and know what it is worth, and a producer who can design a study you can actually finish.”

One of 28 self-contained course books in the MSc Marriage and Family Therapy lecture library: a full taught semester per volume, with cases, practicals, assessment briefs, glossaries and model answers.

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MFT 705 — Research Methods in Marriage and Family Therapy

MSc Marriage and Family Therapy · Semester One · 1.71 Units · 14 teaching weeks · Prerequisite: MFT 701

This volume is one of 28 course books that together cover the whole MSc Marriage and Family Therapy taught programme. Each book is written to carry a 14–16 week semester on its own: every week pairs a core idea with the practice that follows from it, a worked case, and a reflective task, and each book closes with an integration chapter, a glossary and a self-assessment paper with model answers.

The books are designed to be read straight through before the semester begins, revisited week by week, and used again in revision. Cross-references point to the other volumes where a topic is treated in more depth.

How to cite this volume

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A note on the case material

All families, couples and individuals in this book are fictional composites. They are written to teach pattern recognition, not to describe any real person. Nothing here replaces supervised clinical practice, professional registration, legal advice or medical care; legal content is orientation only and must be checked against the law of the relevant jurisdiction.

The lecture library

Twenty-eight self-contained course books carry the taught programme: eight in Semester One, seven in Semester Two, seven in Semester Three including the thesis, and six electives. The volume you are holding is marked in the margin.

Code	Course	Semester	Units
MFT 701	Fundamentals of Marriage and Family Therapy	One	1.71
MFT 702	Theories of Family Therapy	One	1.71
MFT 703	Human Development Across the Lifespan	One	1.71
MFT 704	Ethical and Legal Issues in Therapy	One	1.71
MFT 705	**Research Methods in Marriage and Family Therapy**	One	1.71
MFT 706	Clinical Practice and Techniques	One	1.71
MFT 707	Assessment and Diagnosis in Family Therapy	One	1.71
MFT 708	Cultural Competence in Therapy	One	1.71
MFT 709	Group Therapy Techniques	Two	1.71
MFT 710	Couples Therapy	Two	1.71
MFT 711	Child and Adolescent Therapy	Two	1.71
MFT 712	Crisis Intervention Strategies	Two	1.71
MFT 713	Substance Abuse and Family Dynamics	Two	1.71
MFT 714	Trauma-Informed Care	Two	1.71
MFT 715	Family Systems Theory	Two	1.71
MFT 716	Communication Skills in Therapy	Three	1.71
MFT 717	Psychopathology in Families	Three	1.71
MFT 718	Intimacy and Relationship Dynamics	Three	1.71
MFT 719	Family Life Cycle and Transitions	Three	1.71
MFT 720	Supervision and Consultation in Therapy	Three	1.71
MFT 721	Research Methodology	Three	1.71
MFT 722	Professional Development and Self-Care	Electives	1.71
MFT 723	Special Populations in Therapy	Electives	1.71
MFT 724	Family Law and Advocacy	Electives	1.71
MFT 725	Integrative Approaches in Therapy	Electives	1.71
MFT 726	Neurobiology of Relationships	Electives	1.71
MFT 727	Play Therapy Techniques	Electives	1.71
MFT 770	Master Thesis	Three	1.71

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Course leader's summary. You will become a consumer who can read a paper in ten minutes and know what it is worth, and a producer who can design a study you can actually finish. The course is deliberately practical: every week you do something small with real or simulated data.

How to use this book

Research methods are learned by doing badly, then better. Each week has a **concept**, a **worked example**, and a **small task** you can complete in an hour. The tasks build towards your thesis proposal in MFT 770, so keep every one of them.

1. Course description

Family therapy research is harder than individual therapy research for a structural reason: **people in a family are not independent observations**. If you treat a couple as two independent people, you violate the central assumption of most statistical tests. This course teaches you to see that problem and to handle it in ways that survive examiners and journal reviewers.

The course covers: framing a good question; quantitative, qualitative and mixed designs; sampling and measurement; basic analysis; appraisal of evidence; ethics; and writing a proposal.

2. Learning outcomes

#	Outcome	Assessed
LO1	Convert a clinical hunch into a focused, researchable question	Assignment 1
LO2	Select and justify a design appropriate to the question	Exam, proposal
LO3	Evaluate the reliability and validity of commonly used family measures	Exam
LO4	Handle non-independence and unit-of-analysis problems	Exam, proposal
LO5	Interpret basic descriptive and inferential statistics and effect sizes	Exam, data exercise
LO6	Appraise quantitative, qualitative and mixed-methods studies	Appraisal assignment
LO7	Write an ethical, feasible research proposal	Proposal

3. Assessment

Component	Weight	Due
Question and design exercise (1,000 words)	15%	Week 4
Systematic appraisal of two studies (1,500 words)	25%	Week 9

Component	Weight	Due
Data analysis exercise (set problems)	20%	Week 12
Research proposal (2,500 words)	40%	Week 15

4. The fourteen-week plan

Week	Topic	Task
1	What research is for; evidence-based practice and its critics	Find one paper you will use in the thesis
2	From hunch to question: PICO/SPIDER; feasibility	Draft three questions
3	Designs overview: quantitative, qualitative, mixed	Map your question to a design
4	Sampling: who is in, who is out, who never comes	Critique a convenience sample
5	Measurement: reliability, validity, responsiveness	Appraise one couple measure
6	The independence problem: unit of analysis in family data	Worked analysis of couple data
7	Quantitative analysis I: descriptive and comparison	Run descriptives in software
8	Quantitative analysis II: association, prediction, effect size	Interpret a regression table
9	Qualitative methods and rigour	Code a transcript extract
10	Mixed methods; participatory approaches	Design a mixed study
11	Systematic review and critical appraisal	Appraise one meta-analysis
12	Data management, security, documentation	Write a data plan
13	Research ethics and with vulnerable families	Complete an ethics application
14	Writing it up; publication and impact	Proposal presentations

Week 1 — What research is for

Three legitimate purposes. Description (what is happening), explanation (why, and under what conditions), and improvement (does this intervention help, for whom). A study that serves none of these is a filing cabinet exercise.

Evidence-based practice is the integration of best available research, clinical expertise, and client values and context. It is not "follow the manual". Two critiques worth knowing:

1. **Efficacy versus effectiveness.** Trials conducted with selected volunteers under ideal conditions may not generalize to your clinic, where families present with debts, migration problems, and court dates.
2. **The evidence hierarchy is question-dependent.** A meta-analysis of trials answers "does it work on average?"; a qualitative study answers "how does it work, and for whom does it fail?". Family therapy needs both.

> **TASK**
> Find one paper relevant to your intended thesis topic. Write its citation and one sentence stating what it establishes and one sentence stating what it leaves open.

Week 2 — From hunch to question

A weak question: "Does family therapy work?"

A workable question: "Among adolescents referred to a Lagos community clinic for school refusal, does a six-session family-based intervention reduce days absent from school at 12 weeks, compared with usual care?"

Frameworks. For quantitative work use **PICO** — Population, Intervention, Comparison, Outcome. For qualitative work use **SPIDER** — Sample, Phenomenon of interest, Design, Evaluation, Research type.

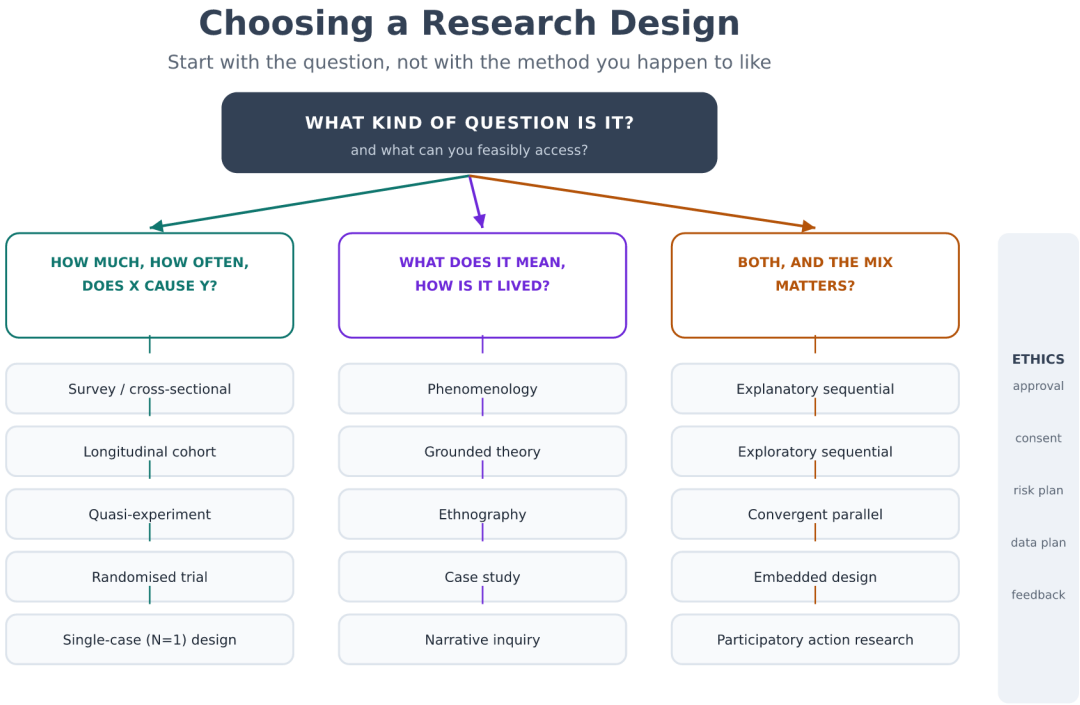
Feasibility screen. Before you fall in love with a design, ask: Do I have access? How many participants can I realistically recruit in six months? Who will approve this? What is the cost? Is the timeline compatible with the semester? Most failed theses are failed at this step, not at analysis.

Common traps.

Trap	Example	Fix
Too broad	"Family functioning in Nigeria"	Specify population, setting, outcome
Unanswerable with your data	"Does family therapy cause lasting change?"	Narrow to what a 12-week design can show
Two questions in one	"Does it work and why?"	One study, one primary question
Question-shaped advocacy	"Why is family therapy better?"	Make it genuinely open ("how does it compare?")

> **TASK**
> Draft three candidate questions. For each, write the PICO or SPIDER elements, the sample you can access, and the main threat to validity.

Week 3 — Designs overview



NOTE

In MFT the unit of analysis is the relationship, not only the individual — which creates real statistical problems: partners are non-independent. Plan for it (dyadic data analysis, actor–partner models, or a design that samples one person per household).

Figure 3.1 — Choosing a research design: start with the question

Quantitative.

Design	Answers	Strengths	Threats
Cross-sectional survey	How common? What is associated?	Cheap, fast	Cannot establish temporality
Longitudinal cohort	What predicts change?	Temporal ordering	Attrition
Quasi-experiment	Does the intervention work without randomisation?	Feasible in clinics	Selection bias
Randomised controlled trial	Does it cause the effect?	Strongest causal inference	Cost, ethics, generalisability
Single-case (N=1) design	Does it work for this family?	Clinically realistic	Limited generalisation

Qualitative.

Approach	Focus	Sample size
Phenomenology	Lived experience of a phenomenon	6–12
Grounded theory	Building a theory from data	15–30, to saturation
Ethnography	Culture and practice in a setting	Prolonged fieldwork
Case study	Depth in one bounded system	1–5 cases

Approach	Focus	Sample size
Narrative inquiry	How people story their lives	3–10
IPA	How individuals make sense of experience	3–6

Mixed methods — see Week 10.

> **TASK**
> Take your strongest question from Week 2 and map it onto two designs. State what each would let you claim, and what it would not.

Week 4 — Sampling

The uncomfortable fact. Clinical samples are almost never representative. Families who attend are different from families who are referred and do not attend, and those are different again from families who are never referred. Your findings describe attenders, not families.

Probability sampling (random, stratified, cluster) permits generalisation. **Non-probability sampling** (convenience, purposive, snowball, quota) does not — and this is fine, if you say so.

In family research specifically:

- **Who is the participant?** Mother, father, child, whole family, or the therapist's rating of the family?
- **Who refuses?** Fathers are the most common absence in family research and in family therapy. A study that reports "family functioning" from mothers' reports is a study of mothers.
- **Sample size.** For quantitative work, do a power calculation before you collect (software: G*Power, or the pwr package in R). For qualitative work, plan for saturation and state how you will judge it.

> **TASK**
> Critique the sample in any published Nigerian family study: who was included, who was excluded, and what does the exclusion do to the conclusions?

Week 5 — Measurement

A measure must be **reliable** (consistent), **valid** (measuring what it claims), and **responsive** (sensitive to change).



Figure 5.1 — Four validity targets and three reliability methods

The four validity targets.

Validity	The question
Internal	Did the intervention, and not something else, produce the change?
External	Can these findings travel to other families, settings and cultures?
Construct	Are you measuring "alliance", or "wanting to please the therapist"?
Statistical	Is the effect large enough, and the test powerful enough, to trust?

Instruments you should know (and check for local validation):

Instrument	Measures	Notes
Dyadic Adjustment Scale (DAS) / DAS-4	Relationship satisfaction	Widely used; short forms available
Revised Dyadic Adjustment Scale (RDAS)	Satisfaction	14 items
FACES-IV	Cohesion and flexibility	Psychometrics debated; use cautiously
Family Assessment Device (FAD)	Family functioning (McMaster)	60 items; subscales
Working Alliance Inventory (WAI-SR)	Therapeutic alliance	Adapt for family work carefully
PHQ-9 / GAD-7	Depression / anxiety	Free, brief, widely validated

Instrument	Measures	Notes
SDQ	Child emotional and behavioural functioning	Parent, teacher, self versions

The cultural validity question. Was this instrument developed and validated with families like the ones you see? If not, report that limitation and consider a local adaptation study (a perfectly good thesis).

> **TASK**
> Take one couple measure. Find two published validations, one in an African sample if it exists. Report: internal consistency, factor structure, and any evidence of measurement invariance.

Week 6 — The independence problem

This is the technical heart of family research.

The problem. In a study of 100 couples, you do not have 100 independent observations; you have 50 couples, and partners' scores are correlated. Standard tests (t-test, ANOVA, OLS regression) assume independence. Violating it inflates Type I error: you will find effects that are not there.

Three solutions.

1. **Analyse at the dyad level.** Compute a couple score (mean, difference, or both) and analyse that. Simple, loses information about within-couple differences.
2. **Treat the dyad as the unit and model non-independence explicitly.** Multilevel models (HLM) with individuals nested in couples nested in therapists. This is the modern standard.
3. **Dyadic models.** The **Actor–Partner Interdependence Model (APIM)** separates actor effects (my predictor on my outcome) from partner effects (my predictor on my partner's outcome). This is often the clinically interesting question.

Worked example. Suppose you measure criticism (predictor) and satisfaction (outcome) in 60 couples. An OLS regression treating all 120 people as independent finds criticism predicts satisfaction, $p = .001$. An APIM re-analysis finds: actor effect significant; partner effect smaller but present; and the standard errors widen. The clinical interpretation changes: criticism damages the person who gives it as well as the person who receives it.

> **TASK**
> Simulate it: in a spreadsheet, create 40 couples with correlated scores. Run a t-test treating all 80 as independent, then compute the couple means and run the test again. Note the difference in p.

Week 7 — Quantitative analysis I

Descriptive statistics. Mean and standard deviation for symmetric data; median and interquartile range for skewed data (income, session counts). Always report n, missingness, and the range.

Comparing groups. t-test (two groups), ANOVA (three or more), chi-square (categorical). Check assumptions: independence, normality (roughly), equal variance.

Clinical significance versus statistical significance. A drop of 2 points on a satisfaction scale can be statistically significant with n = 400 and clinically meaningless. Report **effect sizes** and, where available, the **minimal clinically important difference**.

Common effect sizes.

Statistic	Meaning	Rough benchmark
Cohen's d	Standardised mean difference	0.2 small, 0.5 moderate, 0.8 large
Pearson's r	Association	0.1 small, 0.3 moderate, 0.5 large
Odds ratio	Relative odds	1.5 modest, 2+ substantial
Number needed to treat	How many treated for one extra success	Lower is better

> **TASK**
> Interpret this sentence for a non-researcher in two sentences: "The intervention group showed a greater reduction in distress than controls (d = 0.55, 95% CI [0.21, 0.89], p = .002)."

Week 8 — Quantitative analysis II

Correlation is not causation, and neither is regression. Regression tells you how much of the variance in an outcome is associated with predictors, adjusted for other predictors. It does not tell you the direction, unless the design does.

Reading a regression table. Look at: which predictors are significant; the sign and size of the coefficient (in meaningful units); the confidence intervals; R^2 (variance explained); and whether the model diagnostics were reported.

Moderation and mediation. Moderation: for whom, or under what conditions, does the effect hold? (e.g. does the intervention work better for younger adolescents?) Mediation: through what mechanism? (e.g. does improved communication explain the reduction in distress?). Mediation requires strong assumptions and, ideally, longitudinal data.

> **TASK**
> Take a published regression table from a family study. Write four sentences a clinician could use, including one caveat.

Week 9 — Qualitative methods and rigour

When qualitative methods are the right choice: when you need to know how something is experienced, why a service is not used, what a concept means locally, or how a therapy unfolds. They are not a fallback for people who dislike numbers.

Rigour criteria (Lincoln & Guba; Braun & Clarke for reflexive TA):

Criterion	Practical meaning
Credibility	Do the findings ring true to participants? (member checking, triangulation)
Transferability	Enough context for a reader to judge relevance elsewhere (thick description)
Dependability	An audit trail of decisions
Confirmability	Findings grounded in data, not in the researcher's preferences
Reflexivity	Your own position examined, not hidden

Coding, briefly. Familiarise → code → generate themes → review → define → write. Coding can be inductive (from the data) or deductive (from a framework). Reflexive thematic analysis explicitly treats themes as constructed by the analyst, not excavated from the data.

Sample extracts. Suppose a father says: "They asked me to come, but nobody asked me what I thought was wrong. I sat there like a visitor in my own house." Codes: excluded, not consulted, home as foreign. Candidate theme: "A visitor in my own house": the disengagement of fathers in family therapy. Note how the theme uses the participant's own language.

> **TASK**
> Code the extract above independently, then compare with a peer. Where you disagree, ask whether you are coding the same unit.

Week 10 — Mixed methods and participatory approaches

Why mix. Numbers answer "how much"; words answer "how" and "why". Three designs:

1. **Explanatory sequential** — quantitative first, then qualitative to explain the result.
2. **Exploratory sequential** — qualitative first (to build a culturally valid measure or model), then quantitative.
3. **Convergent parallel** — both at once, integrated in interpretation.

Integration is the hard part. A study that reports a survey and some interviews side by side, without integrating the findings, is two studies in one paper. Use a **joint display** table: finding from strand A

| finding from strand B | convergence / divergence | interpretation.

Participatory action research is particularly well suited to community and family work: families help define the question, collect data, and interpret findings. It strengthens validity and ethics at once.

> **TASK**
> Design a mixed-methods study to understand why families drop out after the first session. Specify the sequence, the integration point, and one joint display row.

Week 11 — Systematic review and critical appraisal

Levels of evidence (roughly, for treatment questions): systematic reviews and meta-analyses > RCTs > quasi-experiments > cohort > case-control > cross-sectional > case series > expert opinion. For questions of meaning, the hierarchy inverts.

Appraising a meta-analysis. Ask: Was the search systematic and pre-registered? Were studies comparable enough to pool? Was heterogeneity (I^2) reported and explained? Was publication bias assessed (funnel plot, Egger's test)? Do the authors' conclusions follow from the data?

A ten-minute paper reading routine.

1. Abstract: design, n, outcome, conclusion.
2. Participants: who, where, how recruited.
3. Comparison: what was the control?
4. Measure: does it measure something you care about?
5. Effect: size and confidence interval, not just p.
6. Limitations: what the authors admit and what they do not.
7. Relevance: your population, your setting, your resources.

> **TASK**
> Appraise one meta-analysis of couple or family therapy. Write 500 words: what it establishes, what it cannot establish, and one clinical implication.

Week 12 — Data management

Principles. Collect the minimum you need; store it encrypted; separate identifiers from data; define retention and destruction; document every decision.

Practical protocol for a master's thesis.

Element	Rule
Identifiers	Separate key file, encrypted, access limited to you and supervisor
Audio/video	Encrypted drive, deletion date agreed with participants

Element	Rule
Transcription	Pseudonymised at the point of transcription, not after
Version control	Date every dataset; never overwrite raw data
Analysis log	One page per week: what you did, what you decided, why
Backup	Two locations, one offline

> **TASK**
> Write a one-page data management plan for your proposed study.

Week 13 — Research ethics

Core requirements. Institutional ethics approval before recruitment; informed consent appropriate to age and capacity; voluntariness (especially for your own clients); protection of confidentiality; a plan for distress and incidental findings; fair selection of participants; and a dissemination plan that returns something to the community.

Specific issues in family research.

- **Consent within families:** if a parent consents, does the adolescent have a genuine choice? Use separate assent and consent processes.
- **Coercion by proxy:** recruiting your own clients creates pressure. Use a third party for recruitment.
- **Group confidentiality:** you cannot guarantee that family members will not repeat what is said.
- **Power and payment:** paying participants can be appropriate compensation or an undue inducement; be transparent about the reasoning.

> **TASK**
> Complete a mock ethics application for your proposal, including consent forms for an adult and for a 14-year-old.

Week 14 — Writing it up

Structure (IMRaD). Introduction (why this question matters; what is known; the gap) → Method (what you did, in enough detail to replicate) → Results (what you found, without interpretation) → Discussion (what it means, limitations, implications) → Conclusion.

Writing rules that examiners reward. One idea per paragraph; active voice where possible; report numbers with units and confidence intervals; state limitations honestly and specifically; do not overclaim in the abstract.

Publication. Target a journal that publishes practitioner research; follow the author guidelines exactly; expect rejection once and revise.

Glossary

Actor–partner interdependence model (APIM) — a dyadic model separating one's own and one's partner's effects. **Attrition** — loss of participants over time. **Confidence interval** — the range within which the true value is likely to fall. **Construct validity** — the degree to which a measure captures the intended concept. **Effect size** — the magnitude of an effect, independent of sample size. **Heterogeneity (I^2)** — the proportion of variation across studies due to real differences rather than chance. **Independence assumption** — the statistical requirement that observations are not correlated. **Measurement invariance** — whether an instrument measures the same construct across groups. **Power** — the probability of detecting an effect that exists. **Saturation** — the point at which new data yield no new themes. **Triangulation** — using multiple methods, sources or analysts to strengthen inference.

Self-assessment

Multiple choice

1. Treating both partners in 50 couples as 100 independent observations primarily threatens: (a) external validity (b) statistical conclusion validity (c) construct validity (d) reliability

2. The APIM distinguishes: (a) pretest and posttest (b) actor and partner effects (c) randomisation and control (d) qualitative and quantitative

3. A qualitative study of lived experience with 8 participants is most likely: (a) grounded theory (b) phenomenology (c) ethnography (d) RCT

4. Cronbach's alpha estimates: (a) test–retest reliability (b) inter-rater reliability (c) internal consistency (d) construct validity

5. The strongest design for causal inference is: (a) cross-sectional survey (b) quasi-experiment (c) randomised controlled trial (d) case series

6. "For whom and under what conditions does the effect hold?" refers to: (a) mediation (b) moderation (c) randomisation (d) sampling

7. Which is not a qualitative rigour criterion? (a) credibility (b) transferability (c) generalisability (d) reflexivity

8. A 95% confidence interval that includes zero suggests: (a) a large effect (b) a non-significant effect at the .05 level (c) a reliable measure (d) a sampling error in the measure

9. The most common absence in family research samples is: (a) mothers (b) fathers (c) adolescents (d) grandparents

10. Reflexivity in qualitative research means: (a) the researcher examines their own position and effect (b) participants review findings (c) the study is replicated (d) data are triangulated

Answer key

1 b · 2 b · 3 b · 4 c · 5 c · 6 b · 7 c · 8 b · 9 b · 10 a

Short answer

1. Convert "does family therapy help adolescents?" into a PICO question.
2. Explain the unit-of-analysis problem in one paragraph a clinician would understand.
3. Name three threats to internal validity in a pre-post clinic study.
4. What is measurement invariance and why does it matter in cross-cultural family research?
5. List five elements of a data management plan.

Essay (choose one, 2,000 words)

1. "The randomised controlled trial is the wrong gold standard for family therapy." Discuss.
2. Design a mixed-methods study of therapy drop-out in a Nigerian family clinic, justifying every methodological choice.
3. Critically appraise the evidence that common factors explain most psychotherapy outcome.

Further reading

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Next: MFT 706 — Clinical Practice and Techniques